# Download the Candidate Data Package

The full assignment package is provided as a ZIP file. Download and unzip it before starting the prework.

## Google Drive Link

https://drive.google.com/file/d/1cEEo7VHYYxNkzf-_HMdNJeGnq8SOOykT/view?usp=drivesdk

## Steps

- Open the Google Drive link above.
- Click Download to save the ZIP file.
- Unzip the downloaded package on your computer.
- Open the assignment brief PDF first, then review the primary PBL data and grant reporting evidence folders.

## Important

All data, grant labels, school codes, geography labels, finance units, writeups, images, and media records are synthetic and for assessment use only.

# Lead Full-Stack Product Developer Assessment

PBL Program Intelligence & Grant Reporting Assistant — Pre-Interview Assignment

## Business Context

Mantra4Change works with education systems across districts, blocks, schools, and program teams. For this assignment, assume the team has collected three months of synthetic Project-Based Learning (PBL) implementation data from schools in a focus geography.

The product should help program and leadership users move from raw response data to review-ready decisions, and from computed evidence to grant-ready reporting.

**This is not a generic dashboard or chatbot task.** We are looking for senior judgment across data modeling, analytics, workflow design, engineering quality, and responsible AI use.

## What Program Intelligence Means Here

Program intelligence is the deterministic layer that converts school-level implementation data into decision support for monthly reviews. It should make progress, gaps, risk, and priorities visible without depending on AI.

- **Ingest and normalize:** combine July, August, and September CSV files into a usable application model.
- **Calculate:** compute participation, evidence submission, enrollment, attendance, and month-over-month movement.
- **Classify:** apply consistent code-based risk thresholds to districts, blocks, and indicators.
- **Explain:** show why a geography or indicator is on track, behind, at risk, or critical.
- **Prioritize:** identify which districts, blocks, grades, or subjects need follow-up first.
- **Prepare review:** generate structured discussion points for a program review meeting.

## Data Provided

The candidate package contains two data groups.

- **Primary PBL data:** three monthly CSV files for July 2025, August 2025, and September 2025. These include school response fields, synthetic school codes, district/block labels, completion, evidence submission, classes, subjects, enrollment, attendance, and derived risk fields.
- **Grant reporting evidence:** three CSV files covering grant profile and finance, grant performance and report material, and evidence/media assets. These are synthetic and intentionally anonymized.

All data is synthetic and for assessment use only. Donor labels, school codes, geography labels, finance units, writeups, images, and media records must not be treated as real records.

## Core User Workflows

| Workflow | Primary User Need | Expected Product Behavior |
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| Program review | Understand progress and risks across months, districts, blocks, grades, and subjects. | Interactive filters, KPI cards, trend views, district/block comparisons, and deterministic risk explanations. |
| Review preparation | Prepare leadership discussion points before monthly review meetings. | Structured summary of achievements, gaps, priority geographies, and discussion prompts. |
| Grant reporting | Create a report-ready grant section using finance, outcomes, milestones, and evidence assets. | Select grant/month, show facts, link evidence, and generate report text that cites computed data. |

## Scope & Prioritization

Build a working end-to-end application for the core program review workflow. A smaller set of complete, well-reasoned features scores higher than many unfinished ones.

### Tier 1: Must Work End-to-End

1. **Program Review Filters:** filter by reporting month, district, block, grade, and subject. Dashboard metrics must update from the selected filters.
2. **Monthly Review Dashboard:** show total schools, participating schools, participation percentage, evidence-submission percentage, total enrollment, attendance, and attendance percentage. Include month-over-month movement for at least two metrics.
3. **District and Block Performance:** show high-performing and low-performing geographies. Users should quickly see where follow-up is needed.
4. **Deterministic Risk & Gap Engine:** classify indicators or geographies using code, not AI. Suggested thresholds: On Track >= 75%; Behind 60% to below 75%; At Risk 35% to below 60%; Critical below 35%.
5. **Grant Reporting Assistant:** let a user select a grant label and reporting month. Show finance utilization, outcomes, milestones, risks, linked evidence assets, and synthetic media records. Generate a report-ready grant section.

### Tier 2: Complete at Least One Enhancement

1. **Monthly Review Summary:** generate a structured program review summary from deterministic insights. Cover achievements, month-over-month changes, risks, priority districts/blocks, and discussion points.
2. **Program Reporting Assistant:** let a user select a month, district, or block and generate a concise report-ready narrative. Include the facts used to produce the narrative.
3. **Report Export:** provide a simple export or copy-ready view for program summaries or grant report sections.

### Tier 3: Optional but Valuable

- Recommended actions: generate or display 3-5 next actions linked to specific district, block, grade, subject, or grant-reporting gaps.
- Each action should include owner, priority, due date, status, and linked metric. These action records may be generated by your application.

## Grant Report Creation: What We Expect

The grant-reporting workflow should not be a generic text box. It should assemble a report section from structured facts, finance rows, milestone status, performance metrics, and evidence assets.

- **Selection:** user selects a grant label and reporting month.
- **Fact panel:** app shows performance metrics, utilization, milestone summary, risk status, and linked evidence/media records.
- **Narrative generation:** AI or mock logic writes a report-ready section using only computed facts and selected evidence.
- **Traceability:** generated text should show the source facts used, such as completion rate, evidence rate, attendance rate, utilization, milestones, and image/media references.
- **Guardrail:** the report should still work as a deterministic fact summary if AI is disabled.

### Acceptable Report Output

- A report preview inside the app is enough.
- Export to PDF or DOCX is optional.
- The narrative should distinguish computed facts from generated explanation.
- Candidates should avoid hallucinated achievements, locations, or evidence not present in the provided data.

## AI Usage Expectations

AI may be used freely during development. Runtime AI use is optional.

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| **Preferred runtime flow:** deterministic calculations -> structured insights -> generated narrative |
| **Avoid:** raw CSV rows -> AI analysis -> business logic |

Dashboard metrics, risk classifications, priority gaps, and grant fact summaries must work even if the AI call is disabled.

## Technical Expectations

- Frontend: React, Next.js, Vite, or similar.
- Backend: Node.js, Express, Next.js APIs, serverless APIs, or similar.
- Database: PostgreSQL, Supabase, SQLite, MongoDB, or static seed data with a clear production plan.
- Language: TypeScript preferred; JavaScript acceptable.
- AI: real LLM or clearly explained mock/rule-based generator.

## Submission Requirements

- GitHub repository.
- Working demo URL, if available.

- README with setup instructions, architecture overview, data model, assumptions, risk logic, AI workflow or mock approach, limitations, production-readiness notes, and future improvements.
- Seed data, scripts, or import instructions.

Clarifying questions may be handled during the live review. Please document assumptions and move forward when the data is intentionally ambiguous.

## Live Review

- Walk through your data model and risk rules.
- Explain how dashboard metrics are calculated.
- Show how AI or mock narratives are produced.
- Make a small live code change.
- Demonstrate that deterministic metrics, risk classifications, and grant fact summaries work with AI disabled.

## Evaluation Criteria

| Area | What Will Be Evaluated |
|---|---|
| Product thinking | User-centric workflow, review-meeting usefulness, decision support, and report usability. |
| Engineering quality | Maintainable architecture, data flow, code organization, and ability to explain tradeoffs. |
| Data & analytics | Correct calculations, trend logic, risk classification, and handling of CSV inputs. |
| AI integration | Grounded outputs, clear fact/narrative separation, validation approach, and graceful fallback. |
| Communication | README clarity, assumptions, limitations, and live explanation quality. |
| Production thinking | Scalability, reliability, security awareness, and operational considerations. |

## Success Definition

Success means connecting PBL implementation data -> program intelligence -> review decisions -> report-ready grant communication -> action. We are not looking for the largest implementation. We are looking for a practical, explainable product that helps teams understand progress and act on gaps.